Techniques for Dynamic Visualization of Time-Varying Flood Data

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Abstract

This project aims to develop visualizations for the general public of satellite-based flood data from Copernicus Emergency Management Services, focusing on the floods that impacted Pakistan in September 2022. The goal is to effectively communicate the extent, likelihood and uncertainty to the general public. Methods include literature review, visualization design, and user evaluation with an emphasis on interpretability and accessibility.

Keywords: Flood Visualization, Geospatial Data, Uncertainty Visualization, Public Communication, Human-Centered Design, Pakistan floods (2022), Time-Varying Data

1 Introduction

The Copernicus Emergency Management Services (CEMS) data set provided for this project consists of satellite-observed, time-dependent flood data next to the city of Larkana, in Sindh province (see for details 1) in Pakistan, located in the Indus river basin in September 2022. The floodings were caused by an abnormal series of rains that caused flooding in the period between April and October 2022. As a result, more than 2.1 million people were affected. The casualties were estimated at 1,729 dead and 12,867 injured. As a representative sample of this phenomenon, we will focus only on data from September. As the extent of flooding varies over time, visualizing these changes in a way that is understandable to a wider audience is a particular challenge.

2 Problem statement

The aim of this project is to develop and propose new visualization techniques that can effectively communicate these time-varying flood events, ensuring that they are accessible to the general public. The research question is the following:

How can flood risk be effectively combined and visualized for the general public?

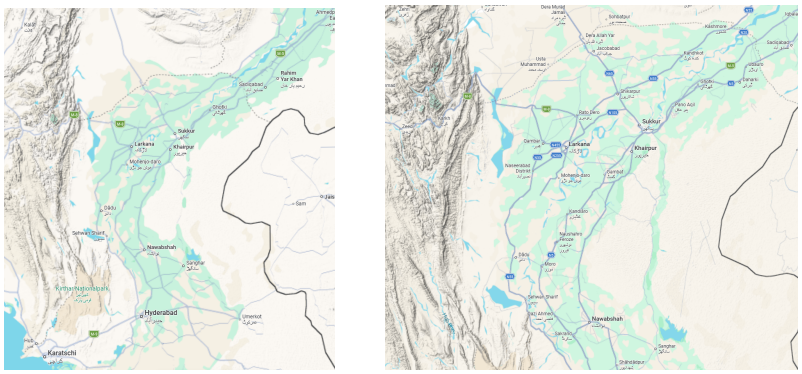


Figure 1: Region of interest: Larkana and Sindh province in Google Maps

We define flood risk as a combination of current and future flood likelihood and flood extent. Our definition of general public refers to civilians without experience in analyzing flood visualizations, who must be able to accurately interpret information regarding flood risks even in unfamiliar parts of the world. For our assessment, we will select users with no formal training in geography, flood monitoring, or data science. Participants will range in age from 18 to 64 years, and the group will include individuals without a university degree. Our target users are not expected to have detailed knowledge of the regions being assessed.

3 Methodology

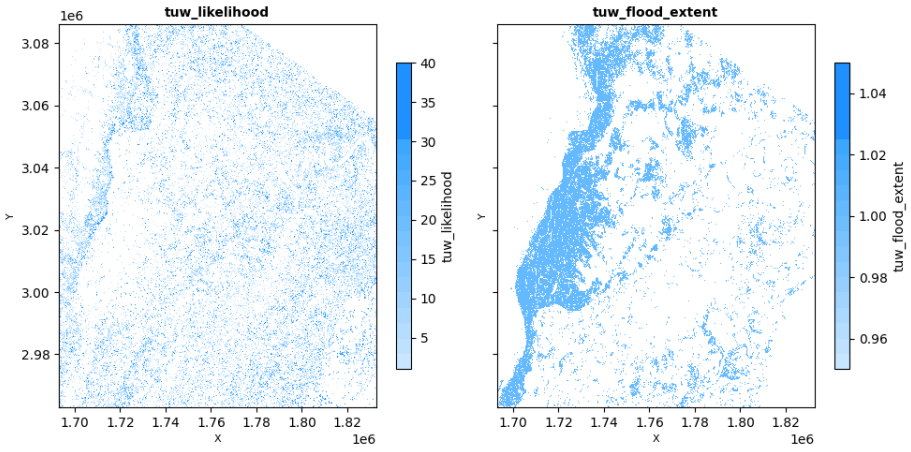


Figure 2: Retrieved parameters flood likelihood and extent provided by CEMS

3.1 Literature Research

To establish a foundation for effective flood visualizations, existing literature on geospatial data representation and interactive mapping tools will be reviewed. Several relevant approaches have emerged in recent years. For instance, Rahman and Thakur [2] propose overlaying flood maps from different timestamps to highlight temporal changes in flood extent. A broader survey by Yussof et al. [4] presents various visualization techniques—including technical, interactive, 3D, and virtual reality approaches—which can significantly improve user engagement with complex geospatial data.

The specific challenge of communicating uncertainty in flood predictions is addressed by both Leedal et al. [1] and Seipel and Lim [3], who emphasize the importance of visual encodings that convey confidence levels without overwhelming the viewer. This review focuses on research published within the last 10 years that specifically addresses the visualization of flood risk (likelihood and extent) for non-expert audiences, ensuring the proposed designs are grounded in proven cartographic and cognitive principles.

3.2 Data

The project uses the Global Flood Monitoring dataset, which provides data for detecting floods from satellite observations. The data is already mostly preprocessed and includes calculated variables (`flood_likelihood` and `flood_extent`; see 2) for selected regions of Pakistan. A mask for the water surface of the permanent body of water is included. Although the Global Flood Monitoring dataset is mostly pre-processed, additional steps would be required to make it suitable for visualizing to the general public. This could include aligning the different variables, such as the flood likelihood, extent, and permanent water bodies.

3.3 Visualization

A new visualization technique will be implemented and compared to the baseline 2 visualization for time-varying flood data. This could include interactive dashboards with time sliders, layered visualizations that combine flood likelihood, extent, and uncertainty, or other methods found in the literature search. The focus lies on developing visualizations that are intuitive, and informative for a broader audience, specifically non-technical users. At least two compelling visualization techniques should be proposed and evaluated against the baseline

3.4 Evaluation

To evaluate the effectiveness of the developed visualizations, a set of qualitative criteria will be applied, including clarity, interpretability, and accessibility. User feedback from performance A/B testing will help assess whether the visualizations improved with regard to the baseline, which are the provided visualizations

of the TU Wien in regards to clarity and interpretability. Accessibility will be assessed with a focus on color blindness, using visual filters and evaluating contrast ratios.

4 Domain Specific Lecture

I successfully completed the Domain-Specific lecture in Data Science, namely *120.110 Data Retrieval in Earth Observation* in 2025S semester.

5 Supervision

Supervisor Dipl.-Ing. Thais Sarinho Beham
Supervisor Dr. Martin Schobben
Co-supervisor Assistant Professor Astrid Weiss

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